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CHECKER

Blockchain Solutions



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Smart Contract Security Audit

WATA TOKEN



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Project Information

Wata is a governance-enabled ERC1363-compatible token deployed on BNB Smart Chain. It features a customizable tax system, blacklisting, role-based access control, and partial ERC1363 functionality. The audit aims to assess the token's contract logic and its behavior in swap scenarios via PancakeSwap.

Token Name	WATA Token
Symbol	WATA
Web Site	https://wataros.io/
Twitter	https://x.com/WatarosDeFi
Telegram	https://t.me/WatarosDeFi
Announcement	https://t.me/WATAROS
Youtube	https://www.youtube.com/@WatarosDeFi
Platform	Binance Smart Chain
Token Type	BEP20
Language	Solidity
Contract Address	0x32E9052De6143CF6C6109c4C85dEF9f0D9b90620
Contract Link	https://bscscan.com/address/0x32E9052De6143CF6C6109c4C85dEF9f0D9b90620
Testnet Link	https://testnet.bscscan.com/address/0xa3be1366be4e9dd2e1a29cc6eb472774e7e24730

Audit Result

 WATA Token has PASSED the smart contract audit with below findings.

(Other unknown security vulnerabilities are not included in the audit responsibility scope)

Audit Result:	PASSED
Ownership:	Actively Owned
KYC Verification:	NA at the date of report edition
Audit Date:	May 02, 2025
Audit Team:	CONTRACTCHECKER

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Findings Summary

Issues Requiring Investor Attention

These findings represent potential risks to token holders due to ownership privileges, even if not currently exploitable in malicious ways:

! Owner can burn any holder's tokens

- The contract owner (DEFAULT_ADMIN_ROLE) can burn tokens from **any address**, without their consent.
- Though intended for administrative flexibility, this function can harm investor balances if misused.

! Blacklisting prevents selling but not buying

- A blacklisted address cannot initiate a transfer (i.e. **cannot sell**).
- However, anyone can still transfer tokens **to** that address (i.e. **buy is allowed**).
- This asymmetry can lead to trapped liquidity.

! Collected taxes go directly to the governor

- No visibility or control over how `_governor` spends tax revenue.
- Taxes are automatically deducted and sent when selling to a SELLING_ROLE-assigned address.

! Transfer proceeds even when contract is paused

- Only `burn()` is wrapped with `whenNotPaused`.
- Transfers, buys, and sells **still execute when paused**, undermining emergency halt expectations.

! Owner can whitelist addresses for zero tax

- Certain wallets can be excluded from taxes entirely.
- This creates potential for **unfair advantage in trading** and **price manipulation** if not publicly disclosed.

! Owner can disable selling by setting tax to 100% or more

- When tax ≥ 10000 (i.e. 100%), the sell transaction fails because there's no amount left to transfer after tax.
- This effectively **blocks selling** for all users interacting with SELLING_ROLE-assigned addresses.
- A limit should be imposed to prevent abusive tax settings.

Developer-Level Considerations

These points highlight unused or misconfigured parts of the contract that can either be cleaned up or better secured:

! pause() does not halt trading

- Only affects burn() — all ERC20 transfers, including PancakeSwap trades, continue uninterrupted.
- Consider extending whenNotPaused to _transfer() or _beforeTokenTransfer() logic.

! Unused or non-functional variables

- TAX_TGE and _tge_plus_1_hours are declared but ineffective because tax logic does not distinguish initial period due to both values being 0.

! MINTER_ROLE exists but unused

- No mint() function present. If minting is not intended in the future, this should be removed to prevent confusion or misuse.

! No max cap on tax percentage

- No check prevents tax from being set to extreme values (e.g., 10,000 = 100%).
- When tax ≥ 10000 , selling fails due to full amount being attempted to send to governor.

⚠ ERC1363 Compatibility may vary by dApp

- While functions like `transferAndCall()` and `approveAndCall()` are implemented, third-party dApps may not properly handle these unless explicitly designed to do so.
- Recommend additional integration testing with real-world use cases.

Wata Token Audit Summary

This report summarizes the audit of the Wata token smart contract. The token is an ERC1363-based contract with transfer taxation, role-based access control, pausing capability, blacklisting, and integration with DEX platforms like PancakeSwap.

Findings Summary

Category	Status	Notes
Role-Based Access	✓ Secure	Only admin can manage roles. Minter role not used.
Transfer Controls	✓ Secure	Restrictions work as expected.
Pause Mechanism	⚠ Partial	Only affects burn. Transfers and swaps unaffected.
Tax System	✓ Functional	Tax is correctly calculated and routed to governor.
ERC1363	⚠ Untested	Not explicitly tested during this round.
Blacklisting	✓ Working	Blacklisted addresses can't sell but can buy.
LP / DEX Integration	✓ Working	Taxes apply only when LP has SELLING_ROLE.
Overflow / Edge Cases	✓ Handled	Common edge cases handled correctly.

Issues & Recommendations

1. **Pause mechanism does not block transfers or swaps**
 - *Impact:* Medium
 - *Recommendation:* Expand pause checks into `_transfer` to fully freeze token activity.
 2. **Tax limit edge case**
 - When tax ≥ 10000 (100%), transaction fails due to insufficient amount left.
 - Consider capping or checking tax in `setTax`.
 3. **Governor role can burn tokens from any user**
 - *Impact:* High (centralization risk)
 - *Recommendation:* Require user consent or restrict burn to owner's own balance.
 4. **ERC1363 Features**
 - *Impact:* Unknown
 - *Recommendation:* Ensure `transferAndCall` and related functions are explicitly tested on DEX and custom receiver contracts.
-

Passed Scenarios

- Governor correctly receives tax.
- `FREETAX_ROLE` exempt from tax.
- Transfers to/from LP work as expected.
- Blacklist prevents selling, not buying.
- Burn fails when paused (as intended).
- Governor cannot sell while blacklisted.
- LP with no `SELLING_ROLE` avoids tax.
- No tax when governor address is unset.
- Edge case behavior: `transfer(0)`, transfer to self.

Executive Summary

CONTRACTCHECKER received an application on May 02, 2025, from the project team of WATA Token to perform a thorough smart contract security audit. The objective was to identify any vulnerabilities present in the source code of WATA Token, as well as any dependencies within the contract. The audit process employed a combination of Static Analysis and Manual Review techniques, conducted by our expert team.

The audit primarily focused on the following considerations:

- **Functionality Testing:** The Smart Contract was subjected to rigorous testing to assess if the intended logic was followed consistently throughout the entire process.
- **Line-by-Line Manual Examination:** Our experts meticulously reviewed the code, examining each line in detail to identify any potential issues or vulnerabilities.
- **Live Testing with Multiple Clients:** The Smart Contract underwent live testing using a Testnet environment, involving multiple clients. This allowed for real-world usage scenarios to be simulated and evaluated.
- **Failure Analysis:** The preparations for potential failures were analyzed to determine how the Smart Contract would perform in the event of bugs or vulnerabilities.
- **Library Version Analysis:** The versions of all libraries utilized in the code were scrutinized to ensure they were up to date, reducing the risk of known vulnerabilities.
- **On-chain Data Security Analysis:** The security of on-chain data was thoroughly assessed to identify any weaknesses or potential risks associated with data storage and handling.

Furthermore, as part of the smart contract security audit, CONTRACTCHECKER conducted an in-depth review of the WATA Token project, focusing on various key aspects. These included access control and authorization, input validation and sanitization, gas optimization and efficiency, event logging and error handling, as well as compliance with best practices and standards.

Through these comprehensive auditing procedures, CONTRACTCHECKER aimed to provide a detailed evaluation of the WATA Token's smart contract security, highlighting any vulnerabilities and recommending appropriate measures for mitigation.

Overview

This audit report provides a comprehensive assessment of the overall security of the WATA Token smart contract. ContractChecker, a trusted security auditing firm, has conducted a thorough analysis of the contract, evaluating the system architecture and codebase to identify potential vulnerabilities, exploitations, hacks, and backdoors. The objective of this audit is to ensure the reliability, correctness, and robustness of the WATA Token smart contract.

Applied Methodology

The audit process employed by Contract Checker followed industry-leading practices and methodologies. Our expert team utilized a comprehensive approach, including the following key elements:

- **Code Design Pattern Analysis:** We conducted a thorough analysis of the code design patterns used in the smart contract. This analysis helped identify any design flaws or architectural weaknesses that could potentially impact the contract's security.
- **Line-by-Line Inspection:** Our experienced auditors performed a meticulous review of the smart contract's code, examining each line in detail. This process aimed to identify any coding errors, vulnerabilities, or potential security risks that could compromise the contract's integrity.
- **Unit Testing Phase:** We executed a robust unit testing phase, where specific units of the smart contract were tested individually to ensure their functionality and resilience. This phase helped identify and address any functional issues or discrepancies within the contract.
- **Automated Testing:** Contract Checker employed automated testing techniques to complement the manual review process. Automated tests were designed to simulate various scenarios and interactions with the contract, helping to identify potential vulnerabilities or weaknesses that may not be apparent through manual inspection alone.

These elements were integrated into our methodology to ensure a comprehensive assessment of the smart contract's security. By combining manual inspection, unit testing, and automated testing, we aimed to provide a holistic evaluation of the contract's reliability and correctness.

Security Assessment

During the assessment, Contract Checker scrutinized the smart contract for various security aspects, including but not limited to:

- **Vulnerability Identification:** We conducted a comprehensive scan of the contract codebase to identify any potential vulnerabilities that could expose the contract to unauthorized access, manipulation, or exploitation.
- **Exploitation Analysis:** Our team performed rigorous testing and analysis to assess the contract's resistance to common exploitation techniques, ensuring its robustness against potential attacks.
- **Backdoor Detection:** We meticulously examined the codebase to identify any backdoors or hidden functionalities that could compromise the security and integrity of the smart contract.

Sound Architecture

The smart contract incorporates a robust and efficient architecture. It follows a modular design, separating functionalities into distinct modules for reusability and scalability. Access control mechanisms ensure authorized operations, while optimized data structures minimize storage costs. The event-driven architecture enables real-time communication, and upgradeability allows for future enhancements. The contract's sound architecture reflects best practices, ensuring secure and efficient operations.

Note: While keeping the content concise, it's important to maintain clarity and ensure that essential information is conveyed effectively.

Code Correctness and Quality

The smart contract underwent a comprehensive review of its source code, with a primary focus on accuracy, readability, sections of high complexity, and the quantity and quality of test coverage. Contract Checker conducted a thorough assessment to ensure that the code is error-free, easily understandable, and properly tested. By examining these critical areas, the contract's overall code correctness and quality were evaluated, providing assurance of reliable and robust execution.

Risk Classification

Vulnerabilities are classified in 3 main levels as below based on possible effect to the contract.

High level vulnerability

Vulnerabilities on this level must be fixed immediately as they might lead to fund and data loss and open to manipulation.

Medium level vulnerability

Vulnerabilities on this level also important to fix as they have potential risk of future exploit and manipulation.

Low level vulnerability

Vulnerabilities on this level are minor and may not affect the smart contract execution.

Manual Audit:

In the manual audit phase, our developers conducted a comprehensive line-by-line examination of the code. This meticulous process involved a thorough review of each line to identify any potential issues or vulnerabilities. To further validate the contract's functionality, we utilized Remix IDE's JavaScript VM and Kovan networks for testing. By combining manual code inspection with real-world testing environments, we aimed to ensure the accuracy and effectiveness of the smart contract.

Automated Audit

Remix Compiler Warnings

During the audit, the smart contract was tested using Solidity's compiler in Remix. While conducting the analysis, we found no issues or warnings reported by the compiler. This indicates that the contract's codebase complied with Solidity's syntax and best practices, further ensuring the contract's integrity and reliability.

Important Notice for Investors

As the ContractChecker team, our primary objective is to conduct a comprehensive audit of the contract code to assess its functionality and identify any potential risks embedded within the code.

Before making any investment decisions, it is crucial to consider several factors. These include the ownership status, approach of the project team, marketing strategies, general market conditions, liquidity, token holdings, and other relevant aspects.

Investors should always exercise due diligence by conducting their own research and carefully managing their risk, considering the various factors that can impact the success of a project.

Disclaimer

This report provides a limited overview of our findings based on our analysis, in accordance with industry best practices at the time of this report, regarding cybersecurity vulnerabilities and issues in the smart contract framework and algorithms, as detailed within this report. It is essential for you to read the full report to obtain a comprehensive understanding of our analysis. While we have conducted our analysis and prepared this report to the best of our abilities, it is important to note that you should not solely rely on this report and cannot hold us liable based on its content, production, or lack thereof. It is crucial for you to conduct your own independent investigations before making any decisions. We provide further details and clarification in the following disclaimer, which we advise you to read in its entirety.

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